VI. (25 points) In Fig. A, two thin parallel plates are ideal conductors, gap d, infinite in y. An EM wave between them travels normal to y, along positive x as a travelling wave， with electric field $E = E_0 \sin(2\pi z/\lambda_z)\sin(2\pi x/\lambda_x - \omega t)$. ω is angular frequency; t is time, λ_z, λ_x are unknown.

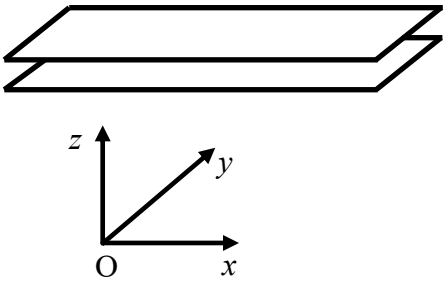


Fig. A

Combining these makes practical microwave transmitting antennas, replacing huge conventional parabolic dishes and greatly reducing antenna cost.

1. Prove that λ_z can take only these values:

$$\lambda_z = 2d/m, \quad m = 1, 2, 3\ldots$$

2. For m=1, find λ_x.

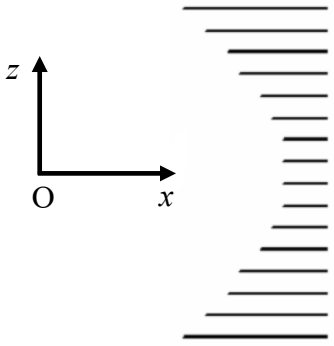


Fig. B

3. Stack equal-gap plates of unequal lengths, parallel and symmetric about an infinite y line source at O. Their right ends align; their lengths vary.
Fig. B is the cross-section in the page. At the right end, a plane EM wave is to emerge along x. Find the equation of the left boundary in the xOz section (Fig. B). (Take m=1; the horizontal distance from the source to the stack's left end is L）。